

Universal k -Fibers of Optimal Compatibility in Periodic Shortest-Path Systems

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Abstract

We prove a universal next-order freedom theorem for optimal compatibility in positive-cost finite-phase periodic shortest-path systems. For every $s \geq k \geq 2$, one fixed completion-total two-phase periodic action system contains a distinguished structural s -simplex and

$$2^{\binom{s}{k}}$$

equal-distance states with one common complete optimal-compatibility truncation through order $k - 1$, while their order- k layers realize every subset of the $\binom{s}{k}$ distinguished k -faces. Equivalently, every k -uniform hypergraph on s vertices occurs as the next-order optimal-compatibility layer inside one common lower-order geodesic-signature fiber. Thus arbitrarily overlapping k -faces can vary independently without a product decomposition, and the next-order pattern count attains the combinatorial maximum on the distinguished simplex.

To formulate this phenomenon, we introduce the k -fiber: for fixed exact distance and fixed optimal-compatibility data through order $k - 1$, it is the set of all realized order- k layers. We show that the finite arithmetic shortest-path compiler extends to an effective k -fiber compiler: every exact next-order pattern has a Presburger-definable realization region and the complete finite fiber is computable. The universal construction is controlled by incidence coordinates h_E : a distinguished prefix J contributes exactly $|J \cap E|$ to h_E , so lower-order prefixes cannot detect the threshold gap between $k - 1$ and k , whereas a k -set reaches the larger threshold only in its own coordinate.

The discovery route begins with shortest-completion geometry in a completion-aware 2:1 deterministic pushdown model, where arithmetic residuals turn completion into a directed shortest-path problem and commuting completion choices appear geometrically. The present article isolates the resulting fiber question at the periodic-system level; all definitions and proofs needed for the main theorem are given here and do not depend on the pushdown model as a hypothesis.

Keywords. periodic shortest paths; optimal compatibility; k -fibers; cubical geodesic compilation; hypergraph realization; Presburger arithmetic; commuting actions.

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1 Introduction

Shortest-path algorithms ordinarily return metric information: an optimal value, one shortest route, or a set of optimal first moves. In structured periodic systems, however, optimality can carry genuinely higher-order local information. Several actions may commute structurally, and each may

occur on a shortest route, while a prescribed set of them may or may not occur jointly as a shortest prefix. The resulting compatibility data form a simplicial object rather than merely a set of optimal edges.

1.1 Motivating origin and discovery route

The present problem arose from shortest-completion geometry in a completion-aware deterministic pushdown automaton for the balance language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

For an input prefix w , the scalar residual

$$\rho(w) = N_1(w) - 2N_0(w)$$

changes by $+1$ on reading a 1 and by -2 on reading a 0. If the current residual is r , a right completion containing x zeros and y ones must satisfy

$$2x - y = r,$$

and shortest completion minimizes $x+y$ over nonnegative solutions. In the completion-aware machine, different shortest orderings of the same minimum demand appear operationally as commuting local choices. This was the concrete origin of the geodesic viewpoint: the stack representation suggests the residual, the residual turns completion into a directed shortest-path problem, and families of shortest completions expose compatibility among commuting actions.

The same line of investigation led from residual completion to a periodic translation model, then to structural cubes and local optimal-compatibility complexes. Two intermediate mechanisms were especially suggestive: a threshold can be arranged so that all compatibility data below a prescribed order remain unchanged while a face at that order changes, and independent products can make several such discrepancies vary separately. The present paper asks what remains when the product mechanism is removed.

The discovery route may be summarized as

$ \begin{aligned} &2:1 \text{ completion-aware DPDA} \longrightarrow \text{arithmetic residual and shortest completion} \\ &\quad \longrightarrow \text{cubical geodesic compatibility} \longrightarrow \text{prescribed-order separation} \\ &\quad \longrightarrow \text{Boolean amplification} \longrightarrow k\text{-fiber universality.} \end{aligned} $
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This diagram records discovery provenance, not logical dependence. The periodic action framework, arithmetic compiler, fiber definitions, and universal construction are developed self-containedly below.

Three design principles from that route are retained in the construction. First, *arithmetic abstraction*: replace a concrete operational representation by coordinates on which shortest distance is explicit. Second, *prescribed-order threshold separation*: arrange thresholds that are invisible below order k but detected exactly at order k . Third, *information separation*: use different coordinates for variable compatibility bits, structural injectivity, distance balancing, and completion, so that changing one layer does not disturb the others.

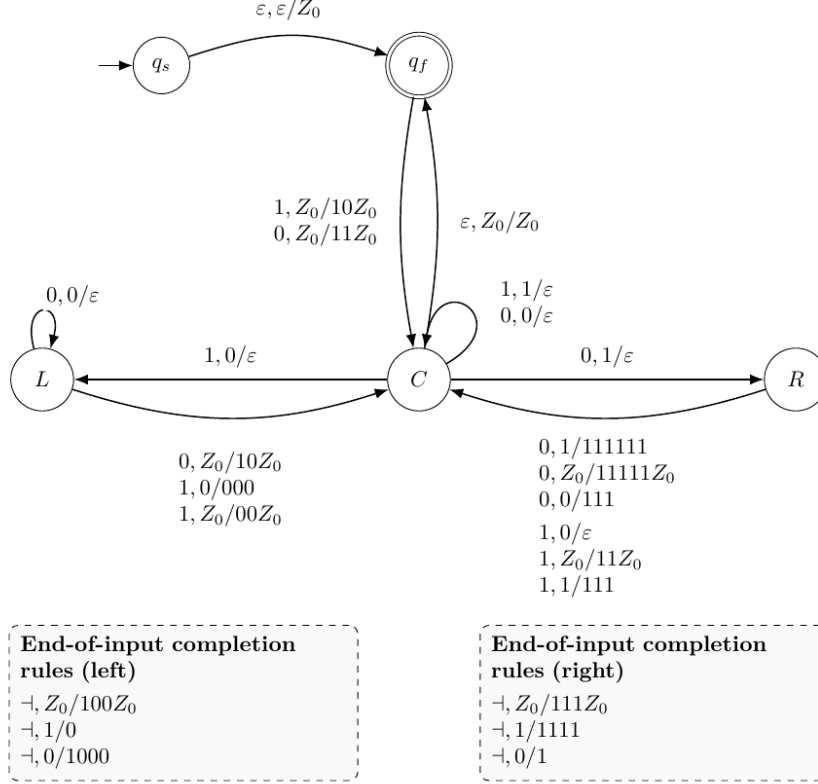


Figure 1: The completion-aware 2:1 deterministic pushdown automaton that supplied the concrete discovery origin. Solid arcs process binary input and the dashed boxes contain right-endmarker completion rules. The automaton motivates the residual, shortest-completion, and commuting-choice viewpoints, but it is not a hypothesis of any theorem in this paper.

1.2 The k -fiber question

For a target-reachable state v , let $H(v)$ denote the local optimal-compatibility complex: its faces are structurally commuting action sets whose every ordering can occur as a shortest prefix. Write

$$H_{\leq h}(v) = \{J \in H(v) : |J| \leq h\}, \quad \Gamma_h(v) = (d(v), H_{\leq h}(v)).$$

Fixing Γ_{k-1} freezes the exact shortest distance and *all* compatibility information through order $k-1$. The central question is then

How much freedom can remain in the next-order layer $H_{=k}$?

We call the collection of all order- k layers realized above one fixed lower-order signature the k -*fiber*.

1.3 Main results

The principal result is the Universal k -Fiber Theorem. For every $s \geq k \geq 2$, we construct one fixed completion-total two-phase periodic action system with s distinguished commuting actions. For each

$$A \subseteq \binom{[s]}{k},$$

there is a state v_A such that all witness states have the same exact distance and the same complete compatibility truncation through order $k - 1$, while

$$J \in H_{=k}(v_A) \iff J \in A \quad (J \in \binom{[s]}{k}).$$

Hence every k -uniform hypergraph on $[s]$ is realized as the complete next-order layer on one fixed structural simplex inside one common lower-order fiber.

The construction is controlled by one incidence coordinate h_E for every potential k -face E . A distinguished prefix J contributes exactly

$$|J \cap E|$$

to h_E . Therefore every prefix of size at most $k - 1$ is insensitive to whether the threshold at h_E is $k - 1$ or k , while a k -set J reaches value k only in its own coordinate h_J ; for every other k -set $E \neq J$,

$$|J \cap E| \leq k - 1.$$

This single intersection inequality is the mechanism that makes arbitrarily overlapping k -faces independent.

The result is extremally sharp on the distinguished simplex. There are exactly $\binom{s}{k}$ possible k -faces and therefore at most $2^{\binom{s}{k}}$ next-order patterns. The construction realizes all of them in one fiber, so the exact distinguished-layer capacity is

$$\text{Cap}_k(s) = 2^{\binom{s}{k}}.$$

A second contribution is algorithmic but subordinate to the universal theorem. The arithmetic shortest-path representation gives a finite k -fiber compiler: for a fixed periodic system and fixed lower-order signature, every exact next-order pattern has a Presburger-definable realization region, and the complete finite fiber is effectively computable.

The contributions can therefore be summarized as follows.

- **k -fiber framework and compilation.** We define next-order fibers of compiled optimal compatibility and show that exact fiber regions and complete fibers are effectively computable in Presburger arithmetic.
- **Universal k -fiber realization.** One common lower-order signature realizes every order- k pattern on a fixed distinguished structural s -simplex.
- **Overlapping-face independence.** The variable k -faces may intersect arbitrarily; their independence is intrinsic to one fiber and does not come from disjoint product factors.
- **Exact next-order capacity.** The universal fiber attains the combinatorial maximum $2^{\binom{s}{k}}$ on the distinguished simplex.

The theorem should not be confused with general realization results for simplicial complexes or squarefree divisor complexes. Cubical state complexes generated by independent local moves are classical in reconfiguration [4], algorithms for shortest paths in metric simplicial complexes study a different ambient-complex problem [5], and squarefree divisor complexes have broad realization phenomena in affine-semigroup theory [6, 8]. Parametric Presburger methods also control complex types in affine-semigroup families [7]. We do not claim novelty for these ingredients in isolation. The

constraint package studied here is different: one fixed shortest-path system, one fixed exact distance, one fixed complete lower-order optimal-compatibility signature, one fixed structural simplex, and every possible next-order pattern on arbitrarily overlapping faces.

The rest of the paper is organized as follows. Section 2 gives a self-contained periodic-action and arithmetic-compiler framework. Section 3 defines k -fibers and proves finite fiber compilation. Sections 4–7 build the universal system and prove the main theorem. Section 8 gives a concrete $s = 3, k = 2$ example, Section 9 proves the exact capacity theorem, and the remaining sections discuss the discovery mechanism, relation to existing realization theories, computational verification, and consequences.

2 Periodic action systems and optimal compatibility

We recall the portion of the cubical geodesic framework needed below. The presentation is self-contained.

2.1 Finite-phase periodic actions

Let Φ be a finite phase set and let I be a finite action alphabet. For each $i \in I$, let

$$\tau_i : \Phi \rightarrow \Phi$$

be a partial phase transition. Whenever $\tau_i(\phi)$ is defined, associate an integral displacement

$$g_i(\phi) \in \mathbb{Z}^n$$

and a positive integer cost

$$c_i(\phi) \in \mathbb{N}_{>0}.$$

The lifted action is

$$T_i(z, \phi) = (z + g_i(\phi), \tau_i(\phi)).$$

Fix a target phase $\phi_\star$ and target state $t = (0, \phi_\star)$. A state v is target-reachable if some finite executable action word carries v to t . For target-reachable v , let $d(v)$ denote minimum path cost to t . When convenient we use the shorthand

$$d(x, \phi) = d((x, \phi)), \quad H(x, \phi) = H((x, \phi)).$$

Multiplying all rational costs by a common denominator preserves the family of shortest paths and scales every distance uniformly, so positive rational costs may be reduced to positive integer costs. Periodic minimum-cost paths with integral gains and rational costs are classical [1].

Proposition 2.1 (Arithmetic reachability and distance). *For every fixed finite-phase periodic action system and every phase ϕ , target reachability and exact distance are effectively Presburger-definable.*

Proof. View each defined phase-action pair as an edge of the finite base phase graph. Let $L_{\phi, \phi_\star}$ be the regular language of base-graph walks from ϕ to the target phase $\phi_\star$. For a walk w , let $G(w) \in \mathbb{Z}^n$ be its total displacement and $C(w) \in \mathbb{N}$ its total cost, and define

$$\mathcal{R}_\phi = \{(G(w), C(w)) : w \in L_{\phi, \phi_\star}\}.$$

The Parikh image of a regular language is effectively semilinear [2]; total displacement and total cost are integral linear functions of the edge-count vector. Effective semilinear sets are closed

under integral linear images and coincide with Presburger-definable sets [3]. Hence $\mathcal{R}_\phi$ is effectively Presburger-definable.

A state (z, ϕ) is target-reachable exactly when

$$\text{Reach}_\phi(z) \iff \exists N \in \mathbb{N} \mathcal{R}_\phi(-z, N).$$

For a target-reachable state, positivity of costs makes the feasible cost set a nonempty subset of $\mathbb{N}$, hence it has a least element. Thus exact distance is defined by

$$\begin{aligned} \text{Dist}_\phi(z, N) \iff & \mathcal{R}_\phi(-z, N) \wedge \\ & \neg \exists N' (0 \leq N' < N \wedge \mathcal{R}_\phi(-z, N')). \end{aligned}$$

This is an effective Presburger formula. □

2.2 Structural cubes

For a phase ϕ and $J \subseteq I$, call J a *structural cube at ϕ* if every permutation of every subset $K \subseteq J$

- (i) is executable from ϕ ;
- (ii) reaches the same final phase;
- (iii) has the same total displacement;
- (iv) has the same total cost;

and distinct subsets $K \neq L$ have distinct lifted endpoints. The family of structural cubes at ϕ is a finite simplicial complex, denoted S_ϕ .

For a structural face $J \in S_\phi$, write $T_J v$ for the common endpoint and $c_J(\phi)$ for the common prefix cost.

2.3 Optimal-compatibility complexes

Definition 2.2 (Optimal-compatibility complex). *Let $v = (z, \phi)$ be target-reachable. For $J \in S_\phi$, define*

$$J \in H(v)$$

if

$$T_J v \text{ is target-reachable} \quad \text{and} \quad c_J(\phi) + d(T_J v) = d(v).$$

For $h \geq 0$, set

$$H_{\leq h}(v) = \{J \in H(v) : |J| \leq h\}, \quad H_{=h}(v) = \{J \in H(v) : |J| = h\}.$$

Because every permutation of a structural cube has the same cost and endpoint, the Bellman equality is equivalent to saying that one, hence every, permutation of J is a shortest prefix. Initial segments of shortest prefixes are shortest, so $H(v)$ is a simplicial subcomplex of S_ϕ .

For fixed system data, membership of a structural face in $H(v)$ is Presburger-definable. Indeed, if $v = (z, \phi)$ and $J \in S_\phi$, then the face guard is

$$\text{Face}_{\phi, J}(z) \iff \exists N, N' \left(\text{Dist}_\phi(z, N) \wedge \text{Dist}_{\tau_J(\phi)}(z + g_J(\phi), N') \wedge N = c_J(\phi) + N' \right).$$

Thus the complete local complex is recoverable from finitely many arithmetic guards.

3 k -fibers and finite fiber compilation

Definition 3.1 (Lower-order signature). *Fix a phase ϕ . For $h \geq 0$ and target-reachable $v = (z, \phi)$, define*

$$\Gamma_h(v) = (d(v), H_{\leq h}(v)).$$

A value $\gamma = (D, K)$ is realizable at phase ϕ if $\Gamma_h(v) = \gamma$ for some target-reachable v in phase ϕ .

Definition 3.2 (k -fiber). *Let $k \geq 1$ and let γ be a realizable order- $(k-1)$ signature at phase ϕ . The k -fiber above γ is*

$$\mathcal{F}_k(\phi, \gamma) = \{H_{=k}(v) : v \text{ is in phase } \phi, \Gamma_{k-1}(v) = \gamma\}.$$

Since S_ϕ is finite, every k -fiber is finite. The point is that it is also effectively computable from the arithmetic compiler.

For $A \subseteq \{J \in S_\phi : |J| = k\}$ define the exact fiber region

$$\Omega_{\phi, \gamma, A}^{(k)} = \{z : \Gamma_{k-1}(z, \phi) = \gamma, H_{=k}(z, \phi) = A\}.$$

Theorem 3.3 (Finite k -fiber compilation). *For every fixed positive-cost finite-phase periodic action system, every phase ϕ , every $k \geq 1$, and every realizable order- $(k-1)$ signature γ :*

- (i) *every region $\Omega_{\phi, \gamma, A}^{(k)}$ is effectively Presburger-definable;*
- (ii) *nonemptiness of every such region is decidable;*
- (iii) *the complete finite fiber $\mathcal{F}_k(\phi, \gamma)$ is effectively computable.*

Proof. The structural complex S_ϕ is finite. Exact equality $\Gamma_{k-1}(z, \phi) = \gamma$ is a finite Boolean combination of the distance predicate and the face guards for all structural faces of size at most $k-1$. Exact equality $H_{=k}(z, \phi) = A$ is a finite conjunction of positive face guards for $J \in A$ and negative face guards for structural k -faces not in A . Hence $\Omega_{\phi, \gamma, A}^{(k)}$ is Presburger-definable. Decidability of Presburger arithmetic determines which of the finitely many candidate patterns have nonempty realization regions. Those and only those patterns form the fiber. $\square$

Remark 3.4. *Theorem 3.3 makes no polynomial-time claim. Its role is structural: for a fixed periodic system the infinite state space admits an exact finite classification of all next-order extensions of any compiled lower-order signature.*

4 The universal construction

Fix integers

$$s \geq k \geq 2.$$

Let

$$[s] = \{1, \dots, s\}, \quad \mathcal{E} = \binom{[s]}{k}, \quad m = |\mathcal{E}| = \binom{s}{k},$$

and put

$$R = \binom{s-1}{k-1}.$$

We construct one fixed two-phase periodic action system $\mathcal{U}_{s,k}$.

4.1 Coordinates and phases

Let the phase set be

$$\Phi = \{G, C\},$$

where G is the geodesic-compatibility phase and C is the completion phase. The target is

$$t = (0, C).$$

Let the coordinate set be

$$Q = \{b\} \sqcup \{p_1, \dots, p_s\} \sqcup \{h_E : E \in \mathcal{E}\}.$$

Thus

$$n = 1 + s + \binom{s}{k}.$$

The coordinate b balances distances among witness states, the coordinates p_i force structural nondegeneracy, and the coordinates h_E encode the next-order compatibility bits.

4.2 Distinguished actions

For each $i \in [s]$, introduce an action

$$a_i : G \rightarrow G$$

with translation

$$u_i = e_{p_i} + \sum_{\substack{E \in \mathcal{E} \\ i \in E}} e_{h_E}$$

and cost

$$c(a_i) = R + 1.$$

The action a_i affects one private coordinate and exactly R incidence coordinates, so

$$\|u_i\|_1 = R + 1 = c(a_i).$$

Write

$$I_\star = \{a_1, \dots, a_s\}.$$

4.3 Structurally isolated completion

Introduce one switch action

$$\sigma : G \rightarrow C$$

with zero translation and cost 1. For every coordinate $q \in Q$, introduce completion actions

$$\xi_q^+ : C \rightarrow C, \quad \xi_q^- : C \rightarrow C$$

with translations $+e_q$ and $-e_q$, respectively, and unit cost. No completion action is executable in phase G , and no distinguished action is executable in phase C .

Lemma 4.1 (Completion-totality). *Every state of $\mathcal{U}_{s,k}$ is target-reachable.*

Proof. From phase C , use coordinate unit actions to reach the origin. From phase G , apply σ and then complete coordinatewise in phase C . $\square$

5 Exact metric and structural isolation

Lemma 5.1 (Completion-phase metric). *For every $x \in \mathbb{Z}^Q$,*

$$d(x, C) = \|x\|_1.$$

Proof. Coordinate unit moves give a path of cost $\|x\|_1$. Conversely, the total cost of every completion path equals the sum of the ℓ_1 -norms of its unit translations, which is at least the norm of the total displacement $-x$ by the triangle inequality. $\square$

Lemma 5.2 (Geodesic-phase metric). *For every $x \in \mathbb{Z}^Q$,*

$$d(x, G) = 1 + \|x\|_1.$$

Proof. Switching immediately to phase C and then completing coordinatewise gives the upper bound.

For the reverse inequality, consider any path from (x, G) to the target. Before its unique first transition to phase C , it performs some word in the distinguished actions. Let U be the total distinguished displacement and C_U its cost. Since each distinguished displacement is coordinatewise nonnegative,

$$C_U = \sum_j \|u_{i_j}\|_1 = \|U\|_1.$$

After the switch, Lemma 5.1 gives remaining cost at least $\|x + U\|_1$. Therefore the total cost is at least

$$\|U\|_1 + 1 + \|x + U\|_1 \geq 1 + \|x\|_1$$

by the triangle inequality. $\square$

Lemma 5.3 (Distinguished structural simplex). *The action set $I_\star$ spans a structural simplex in phase G .*

Proof. The distinguished actions are translations that preserve phase G , so they commute and every ordering of every subset is executable with the same displacement and cost. For a subset $J \subseteq [s]$, the p_i -coordinate of its total displacement is 1 exactly when $i \in J$. Hence distinct subsets have distinct endpoints, proving nondegeneracy. $\square$

Lemma 5.4 (Auxiliary structural isolation). *The structural complex in phase G is*

$$S_G = 2^{I_\star} \cup \{\{\sigma\}\}.$$

In particular, no structural face contains both σ and a distinguished action.

Proof. The singleton $\{\sigma\}$ is structural. If a set contains σ and some a_i , then the permutation that executes σ first cannot execute a_i afterward, because a_i is undefined in phase C . Thus every such mixed set fails the structural-cube condition. No completion action is defined in phase G . $\square$

6 Hypergraph-coded witness states

For each

$$A \subseteq \mathcal{E},$$

let

$$L = m + 1$$

and define $v_A \in \mathbb{Z}^Q$ by

$$\begin{aligned} (v_A)_b &= -(L - |A|), \\ (v_A)_{p_i} &= -1 \quad (i \in [s]), \\ (v_A)_{h_E} &= -(k - 1 + \mathbf{1}_{E \in A}) \quad (E \in \mathcal{E}). \end{aligned}$$

Thus

$$(v_A)_{h_E} = \begin{cases} -k, & E \in A, \\ -(k - 1), & E \notin A. \end{cases}$$

We regard (v_A, G) as the witness state encoding the k -uniform hypergraph $([s], A)$.

Lemma 6.1 (Equal distance). *All witness states have the same exact shortest distance*

$$D_{s,k} = s + 2 + k \binom{s}{k}.$$

Proof. Since $L = m + 1$,

$$\begin{aligned} \|v_A\|_1 &= L - |A| + s + \sum_{E \in \mathcal{E}} (k - 1 + \mathbf{1}_{E \in A}) \\ &= L + s + m(k - 1) \\ &= s + 1 + mk. \end{aligned}$$

Lemma 5.2 gives

$$d(v_A, G) = 1 + \|v_A\|_1 = s + 2 + mk,$$

independent of A . □

For $J \subseteq [s]$, write

$$U_J = \sum_{i \in J} u_i.$$

The crucial incidence identity is

$$(U_J)_{h_E} = |J \cap E| \quad (E \in \mathcal{E}). \tag{1}$$

Also

$$(U_J)_{p_i} = \mathbf{1}_{i \in J}, \quad (U_J)_b = 0.$$

Because all coordinates of U_J are nonnegative,

$$\|U_J\|_1 = |J|(R + 1) = c_J.$$

Lemma 6.2 (Coordinatewise geodesic criterion). *For every $A \subseteq \mathcal{E}$ and every $J \subseteq [s]$,*

$$\{a_i : i \in J\} \in H(v_A, G)$$

if and only if

$$U_J \leq -v_A$$

coordinatewise.

Proof. By Lemma 5.2, the Bellman equality for the distinguished prefix is

$$\|U_J\|_1 + 1 + \|v_A + U_J\|_1 = 1 + \|v_A\|_1.$$

Every coordinate of v_A is nonpositive and every coordinate of U_J is nonnegative. Therefore

$$\|v_A + U_J\|_1 = \|v_A\|_1 - \|U_J\|_1$$

holds exactly when no coordinate overshoots zero, equivalently $U_J \leq -v_A$ coordinatewise. $\square$

7 Universal k -fiber realization

Lemma 7.1 (Complete lower-order invariance). *For every $A, B \subseteq \mathcal{E}$,*

$$H_{\leq k-1}(v_A, G) = H_{\leq k-1}(v_B, G).$$

More precisely, every distinguished subset of cardinality at most $k-1$ is optimal at every witness state, the switch singleton $\{\sigma\}$ is optimal at every witness state, and these are all structural faces of phase G through order $k-1$.

Proof. Let $J \subseteq [s]$ with $|J| \leq k-1$. At a private coordinate p_i ,

$$(U_J)_{p_i} \leq 1 = -(v_A)_{p_i}.$$

At an incidence coordinate h_E , equation (1) gives

$$(U_J)_{h_E} = |J \cap E| \leq |J| \leq k-1 \leq -(v_A)_{h_E}.$$

The balance coordinate receives zero distinguished displacement. Hence $U_J \leq -v_A$ coordinatewise, and Lemma 6.2 makes J optimal for every A .

The switch singleton is optimal because

$$1 + d(v_A, C) = 1 + \|v_A\|_1 = d(v_A, G).$$

By Lemma 5.4, there are no other structural faces in phase G . $\square$

Let

$$K_{s,k} = H_{\leq k-1}(v_A, G), \quad \gamma_{s,k} = (D_{s,k}, K_{s,k});$$

these are independent of A by Lemmas 6.1 and 7.1.

Lemma 7.2 (Exact k -face decoder). *For every $A \subseteq \mathcal{E}$ and every $J \in \mathcal{E}$,*

$$\{a_i : i \in J\} \in H(v_A, G) \iff J \in A.$$

Proof. At the coordinate h_J , equation (1) gives

$$(U_J)_{h_J} = |J| = k.$$

If $J \in A$, then $-(v_A)_{h_J} = k$, so this coordinate is saturated exactly. If $J \notin A$, then $-(v_A)_{h_J} = k - 1$, so the prefix overshoots this coordinate and cannot be geodesic.

It remains to check that no other coordinate obstructs the case $J \in A$. Private coordinates are saturated at most once. For $E \neq J$ with $|E| = |J| = k$,

$$|J \cap E| \leq k - 1 \leq -(v_A)_{h_E}.$$

The balance coordinate is unchanged. Thus Lemma 6.2 gives the equivalence. $\square$

Theorem 7.3 (Universal k -Fiber Theorem). *For every $s \geq k \geq 2$, there exists a fixed completion-total positive-cost two-phase periodic action system $\mathcal{U}_{s,k}$, a phase G , a distinguished structural s -simplex $I_\star$, and a realizable order- $(k - 1)$ signature $\gamma_{s,k}$ whose complete k -fiber is the full power set of the distinguished k -faces:*

$$\boxed{\mathcal{F}_k(G, \gamma_{s,k}) = 2^{\binom{I_\star}{k}}.}$$

Indeed, the witness family

$$\{(v_A, G) : A \subseteq \binom{[s]}{k}\}$$

consists of $2^{\binom{s}{k}}$ equal-distance states with one common complete lower-order compatibility truncation, and

$$H_{=k}(v_A, G) = \{\{a_i : i \in J\} : J \in A\}.$$

Proof. Equal distance and common order- $(k - 1)$ signature follow from Lemmas 6.1 and 7.1. Since $k \geq 2$, Lemma 5.4 implies that every structural k -face in phase G lies entirely in the distinguished simplex. Hence every member of the fiber is a subset of $\binom{I_\star}{k}$. Lemma 7.2 shows that the witness v_A realizes exactly the subset indexed by A . As A ranges over all subsets of $\binom{[s]}{k}$, every subset of $\binom{I_\star}{k}$ occurs. This proves the displayed equality. $\square$

Corollary 7.4 (Uniform-hypergraph realization). *For every $s \geq k \geq 2$, every k -uniform hypergraph $([s], A)$ occurs as the complete order- k optimal-compatibility layer on one fixed distinguished structural simplex, while exact shortest distance and the complete optimal-compatibility structure through order $k - 1$ remain unchanged.*

Corollary 7.5 (Graphs inside one 2-fiber). *For every $s \geq 2$, one fixed periodic shortest-path system has a common order-1 signature whose order-2 fiber realizes every simple graph on s distinguished vertices.*

Corollary 7.6 (Overlapping-face independence). *Let $\mathcal{A} \subseteq \binom{[s]}{k}$ be any family, with arbitrary intersections among its members. Inside one common order- $(k - 1)$ fiber, the membership bits of all faces in $\mathcal{A}$ can be prescribed independently.*

8 A worked universal fiber: $s = 3$, $k = 2$

The smallest nontrivial example already displays the entire phenomenon. Take

$$s = 3, \quad k = 2, \quad \mathcal{E} = \{12, 13, 23\}.$$

Then $m = 3$, $R = 2$, and $L = 4$. The coordinate set is

$$Q = \{b, p_1, p_2, p_3, h_{12}, h_{13}, h_{23}\},$$

and the distinguished actions have translations

$$\begin{aligned} u_1 &= e_{p_1} + e_{h_{12}} + e_{h_{13}}, \\ u_2 &= e_{p_2} + e_{h_{12}} + e_{h_{23}}, \\ u_3 &= e_{p_3} + e_{h_{13}} + e_{h_{23}}, \end{aligned}$$

all of cost 3.

Choose, for example,

$$A = \{12, 23\}.$$

The corresponding witness vector, in the coordinate order displayed above, is

$$v_A = (-2, -1, -1, -1, -2, -1, -2).$$

Its ℓ_1 -norm is 10, so the geodesic-phase distance is

$$d(v_A, G) = 11.$$

The same distance 11 holds for all $2^3 = 8$ choices of $A \subseteq \{12, 13, 23\}$.

Every distinguished singleton is optimal at every witness state, and the switch singleton is also always optimal. Hence the complete common order-1 truncation is

$$H_{\leq 1} = \{\emptyset, \{\sigma\}, \{a_1\}, \{a_2\}, \{a_3\}\}.$$

At order 2, however, the three distinguished pairs are independent. For the present A , the pair 12 contributes 2 to h_{12} and exactly saturates its threshold 2, while 23 similarly saturates h_{23} . Thus

$$\{a_1, a_2\}, \{a_2, a_3\} \in H(v_A, G).$$

The pair 13 contributes 2 to h_{13} , whose threshold is only 1 because $13 \notin A$; it overshoots and therefore

$$\{a_1, a_3\} \notin H(v_A, G).$$

Consequently

$$H_{=2}(v_A, G) = \{\{a_1, a_2\}, \{a_2, a_3\}\}.$$

As A ranges over all subsets of $\{12, 13, 23\}$, the common order-1 signature remains fixed and the order-2 layer runs through all eight simple graphs on the three distinguished vertices. This is the $k = 2$ instance of universal next-order freedom.

9 Exact next-order capacity

The universal theorem attains the obvious combinatorial upper bound once the number of distinguished structural vertices is fixed.

Definition 9.1 (Distinguished k -fiber capacity). *For $s \geq k \geq 2$, define $\text{Cap}_k(s)$ to be the supremum, over all positive-cost finite-phase periodic action systems, phases ϕ , realizable order- $(k-1)$ signatures γ , and distinguished action sets $I_\star \subseteq I$ with $|I_\star| = s$ and $2^{I_\star} \subseteq S_\phi$, of*

$$\left| \left\{ H_{=k}(v) \cap \binom{I_\star}{k} : \Gamma_{k-1}(v) = \gamma \right\} \right|.$$

Theorem 9.2 (Exact distinguished-layer capacity). *For every $s \geq k \geq 2$,*

$$\boxed{\text{Cap}_k(s) = 2^{\binom{s}{k}}}.$$

Proof. An s -vertex simplex has exactly $\binom{s}{k}$ possible k -faces, so no fiber can induce more than $2^{\binom{s}{k}}$ distinct patterns on those faces. Theorem 7.3 realizes every such pattern inside one common fiber, attaining the upper bound. $\square$

Thus the result is not merely an exponential lower bound. Relative to a fixed distinguished structural simplex, lower-order optimal compatibility leaves the next-order layer with the maximum combinatorially possible freedom.

10 Discovery mechanism: from separation to universality

The proof of the Universal k -Fiber Theorem is self-contained, but the route by which the construction was found is informative. It passed through three increasingly strong forms of higher-order freedom.

Prescribed-order threshold separation. The first mechanism is to arrange two nearby threshold values so that every prefix of size at most $k-1$ lies below both thresholds, while a selected k -prefix reaches the larger one and crosses the smaller one. This shows how a compatibility discrepancy can first become visible at an arbitrarily prescribed order. In the present construction this mechanism is retained coordinatewise: the only variable threshold values are $k-1$ and k .

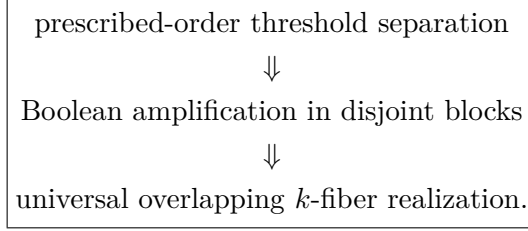
Boolean amplification by independent blocks. A second step is to place several threshold-separation blocks side by side. If the blocks use disjoint action sets, their order- k discrepancies become independent and generate a Boolean family of higher-order signatures over common lower-order data. This demonstrates multiplicity, but the independence is externally supplied by a product decomposition.

Universal overlapping freedom. Theorem 7.3 removes the product mechanism. All variable k -faces lie on one fixed set of s distinguished actions and may overlap arbitrarily. The incidence-coordinate identity

$$(U_J)_{h_E} = |J \cap E|$$

replaces factor independence. The threshold gap between $k-1$ and k makes every proper lower-order prefix insensitive to the encoded bit, while a k -set reaches the larger value only in its own coordinate.

The progression is therefore



The coordinate architecture also separates four logically different tasks. The incidence coordinates h_E carry the variable next-order bits; the private coordinates p_i force structural injectivity; the balance coordinate b keeps all witness distances equal; and the completion phase C makes the system completion-total without creating mixed structural faces. This separation of roles is what allows the order- k layer to vary freely while the entire lower-order signature remains frozen.

11 Relation to existing realization theories

Several ingredients surrounding the present result are classical, and the novelty boundary should be kept narrow.

Periodic shortest paths with integral vector gains and rational costs were studied by Höfting and Wanke [1]. Semilinear and Presburger descriptions arise from classical Parikh-image and Presburger-definability results [2, 3]. Cubical state complexes generated by independent local moves are central in reconfiguration theory, where realization and classification results are known [4]. Shortest-path algorithms on a *given* metric simplicial complex have also been studied [5]; that setting differs from ours, where the simplicial object is itself selected by joint shortest-prefix optimality in an infinite periodic graph. In affine-semigroup theory, squarefree divisor complexes encode divisibility data and admit very broad realization behavior; Bruns and Herzog show, in particular, that every simplicial complex can be realized up to homotopy in suitable semigroup settings [6]. Further constructions realize additional families of squarefree divisor complexes [8], and parametric Presburger arithmetic has been used to control invariants and complex types in affine-semigroup families [7].

Accordingly, the present paper does not claim novelty for simplicial or hypergraph realization in isolation, for commuting actions forming cubes, or for Presburger definability. The structural constraint package is the point: one fixed periodic shortest-path system, one fixed phase, one fixed exact shortest distance, one fixed complete compatibility truncation through order $k - 1$, one fixed structural simplex, and every possible order- k pattern on that simplex. The result concerns freedom *inside one geodesic-signature fiber*, not unconstrained realization across unrelated systems or semigroup elements.

12 Computational verification

The proofs above are symbolic. As an independent regression check, the construction was exhaustively evaluated for several small parameter pairs. For each pair (s, k) , every witness state v_A was generated, exact distance was checked from the closed metric formula, the complete structural truncation through order $k - 1$ in phase G was evaluated, and every distinguished k -face was tested against the Bellman equality.

s	k	$\binom{s}{k}$	witness states	common distance	realized patterns
2	2	1	2	6	2
3	2	3	8	11	8
4	2	6	64	18	64
5	2	10	1024	27	1024
3	3	1	2	8	2
4	3	4	16	18	16
5	3	10	1024	37	1024
4	4	1	2	10	2
5	4	5	32	27	32

In every tested case, all witness states had one common exact distance and one common complete lower-order compatibility truncation, while the distinguished order- k layer ranged over all $2^{\binom{s}{k}}$ prescribed patterns. These computations are not premises of any theorem.

13 Discussion

The k -fiber framework separates two questions that are conflated by statewise shortest-path analysis. The compiler asks which higher-order compatibilities hold at a given state. Fiber theory asks which next-order layers remain possible after all lower-order information and exact distance have already been fixed.

The universal theorem shows that there is no general next-order reconstruction principle even when one records the *entire* lower-order complex. More strongly, the failure is not sparse. On a distinguished s -simplex, all $\binom{s}{k}$ next-order bits can remain independent simultaneously. Thus the ambiguity left by lower-order data can have information content exactly

$$\binom{s}{k}$$

binary bits at order k .

The construction also explains why overlap does not force dependencies. Independence is usually obtained by placing choices in disjoint factors. Here it is obtained arithmetically: each possible k -face has its own incidence coordinate, but actions are shared across coordinates according to the incidence relation. Overlap is therefore present in the action sets themselves, while the intersection inequality separates their geodesic thresholds.

The theorem concerns a fixed distinguished structural simplex rather than the total action count of the full completion-total system. This distinction is deliberate. Completion actions are structurally isolated in a second phase so that they do not contaminate the next-order layer. Determining sharp universal bounds in terms of the *total* number of action labels is a different extremal problem and is not needed for the present exact capacity theorem.

14 Conclusion

We introduced k -fibers for compiled optimal-compatibility signatures in positive-cost finite-phase periodic shortest-path systems. A k -fiber collects all order- k compatibility layers consistent with one fixed exact distance and one fixed complete compatibility truncation through order $k - 1$. The

finite arithmetic compiler extends to this level: exact fiber regions are Presburger-definable and the complete finite fiber is effectively computable.

The main structure theorem shows that such fibers can be universally free. For every $s \geq k \geq 2$, one fixed completion-total periodic system contains a distinguished structural s -simplex and a common lower-order signature whose k -fiber realizes every subset of the $\binom{s}{k}$ distinguished k -faces. Hence every k -uniform hypergraph occurs as the next-order optimal-compatibility layer while exact distance and all lower-order compatibility remain unchanged.

Higher-order independence does not require disjoint factors. Arbitrarily overlapping k -faces can vary independently inside one common geodesic-signature fiber. The resulting pattern count

$$2^{\binom{s}{k}}$$

attains the combinatorial maximum on a distinguished structural s -simplex. In this precise sense, lower-order optimal compatibility places no universal restriction on the next-order layer.

Declarations

Author contributions. Alp Eren Bütün: Conceptualization, Methodology, Formal analysis, Investigation, Computational verification, Writing – original draft, Writing – review & editing. The sole author developed the framework and constructions, proved the mathematical results, carried out the computational regression checks, and prepared the manuscript.

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Data availability. The paper is theoretical and uses no external dataset. Finite instances used for computational regression checks are generated directly from the explicit constructions stated in the manuscript.

Code availability. No software package is required to establish the results. Computational checks were used only as independent regression support and are not premises of any theorem.

Ethics approval and consent. Not applicable. The work is purely mathematical and does not involve human participants, animals, clinical data, or personal data.

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